

Hiring Your Own: Alumni Networks and Inequality in Mutual Fund Manager Hiring

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Abstract

This paper examines whether alumni networks influence hiring decisions in delegated asset management and whether such networks generate inefficiencies in managerial labor markets. Using a comprehensive sample of mutual fund manager appointments, I show that funds are nearly twice as likely to hire a manager who shares an alma mater with the incumbent manager. This hiring preference is primarily driven by undergraduate and MBA networks and is stronger among larger funds, during economic downturns, and when labor market conditions are weaker. Importantly, alumni-connected hires underperform non-connected hires by 1.4 percentage points in the year following the appointment. Consistent with investors recognizing these frictions, funds that hire alumni-connected managers experience lower capital inflows, shorter manager tenures, and lower survival rates. Overall, the evidence suggests that alumni networks shape hiring decisions in asset management and may generate agency frictions that impose costs on investors.

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1 Introduction

Individual managers play a pivotal role in shaping corporate behavior and driving economic performance. Consequently, understanding how companies identify and hire effective managers is crucial to akin to determining their value creation strategies. An emerging strand of literature seeks to answer this by investigating whether and how managers influence firm behavior and performance (Bertrand and Schoar, 2003; Jenter et al., 2016). Recent research indicates that a significant number of newly appointed CEOs have strong affiliations with the hiring board, resulting in connections being important instruments in executive recruitment (Cziraki and Jenter, 2022). One plausible explanation is that employees' networks can benefit companies by providing valuable connections and information (Beaman and Magruder, 2012; Dustmann et al., 2016; Burks et al., 2015; Pallais and Sands, 2016; Brown et al., 2016). The significance of network, particularly educational ones, is evident in the financial sector, as demonstrated by the connections between mutual fund managers and boards of public firms (Cohen et al., 2008), hidden connections across mutual funds and public firms (Ammann et al., 2022), and alumni ties between new venture capital partners and startup founders (Huang, 2022). Yet, the effect of connections between new and existing mutual fund managers on the hiring process remains understudied, as does the efficiency of mutual funds' reliance on such networks.

In this paper, we undertake an empirical analysis of over 14,000 mutual fund manager hiring decisions from 2004 to 2023 using data sourced from MorningStar. While existing research tends to emphasize past performance in manager hiring decisions, our study explores the potential way of alumni ties between incumbent and newly hired fund managers. The core research question posed is: Are mutual funds more likely to hire new fund managers with the same educational background as their existing team? Our hypothesis posits that mutual funds are inclined to hire managers from their alumni network. Furthermore, we investigate the performance ramifications of such hiring preferences: does

fund performance differ based on shared university affiliations among managerial teams?

Addressing these questions entails navigating two primary challenges. Firstly, conventional network evaluation metrics may be limiting. Although a significant portion of existing literature examines information flows between mutual fund managers and executives in public corporations, we contribute a novel perspective by focusing on educational ties among fund managers. Our unique dataset integrates individual educational backgrounds with mutual fund performance metrics, shedding light connections that are inherently ex-ante and less contingent on post-hiring investment outcomes. Such an approach offers valuable insights into mutual funds' networking dynamics.

The second challenge lies in the potential endogeneity when assessing the impact of alumni-based hirings on future manager performance and related attributes. To mitigate this concern, we employ the propensity score matching algorithm by comparing managers with alumni ties to a closely matched set of peers lacking such connections but indistinguishable in terms of observable characteristics at the time of acquiring. By estimating the average treatment effect of treated group, we aim to discern the causal effects of alumni networks on managerial outcomes.

Our initial empirical analysis evaluates the influence of alumni networks on manager hiring decisions. Given that data only reflects actual fund-manager pairings, we adopt a methodology inspired by Hellmann et al. (2008) to approximate a yearly candidate pool based on those potentially considered but ultimately not hired managers. Our Ordinary Least Squares regression estimates confirm the influence of alumni networks in hiring decisions, showing a significant increase of 0.892% in hiring managers with alumni ties after adjusting for fund, year, and manager fixed effects. We further discover that alumni ties from undergraduate and MBA programs are particularly influential in these decisions.

Our exploration also spans the differential effects of networks based on fund size, time periods, and macroeconomic conditions. Firstly, splitting our sample to large funds and

small funds, we see an increased influence of alumni networks on hiring decisions, with a 1.065% rise in hirings from alumni ties. Secondly, the influence of alumni networks has grown over time, being notably stronger post-2014 compared to the period of 2004 to 2013. Third, during challenging economic periods, delineated by NBER Recession index or when the U.S. unemployment rate surpasses 6%, the reliance on alumni networks become even more pronounced.

Finally, we evaluate the efficiency of the mutual fund talent acquisition process. Our findings are revealing: Firstly, managers hired due to alumni ties exhibit a raw return underperformance of 1.38% in the year following their acquisition and 1.55% in the second year. Second, for up to ten years after being hired, managers hired based on alumni connections tend to experience shorter tenures and reduced survival rates. Third, a flow-return sensitivity analysis shows that for every 1% negative return in the hiring year, a manager with alumni connections can attract an additional 3.89% capital to their funds. This suggests that hiring alumni can aid funds in retaining capital even when performance is less satisfactory in the hiring year.

The structure of the paper is as follows: Section 2 review the literature and situates our contribution. Section 3 details our data sources and empirical approach. Section 4 analyzes the implications of alumni ties on managerial outcomes. Section 5 provides robustness checks, and section 6 offers concluding remarks.

2 Literature Review

This paper makes significant contributions to three main bodies of literature. First, this paper contributes to the literature on networks in government bureaus and financial markets. The role of social and educational connections influencing outcomes in governmental and financial contexts has been well-documented. For instance, Fisman et al.

(2020) observed that hometown and the college affiliations can decrease the likelihood of an individual's selection to China's Politburo. Moving from politics to finance, Engelberg et al. (2012) found that banks tend to offer more favorable interest rates to corporations with which they share social ties. In the Venture Capital realm, Hellmann et al. (2008) noted that firms with previous venture capital associations receive preferential loan terms from banks. Hochberg et al. (2007) highlighted that well-networked venture capital investors outperform their peers, while Huang (2022) evidenced that increased investment (but also higher failure rates) in startups by venture capitals with alumni connections. Within the mutual fund sector specifically, networking has been shown to hold significant sway. Cohen et al. (2008) demonstrated that mutual fund managers exhibit asymmetric investment to firms run by executive alumni, leading to more profitability. Ammann et al. (2022) exposed significant abnormal returns stemming from hidden connections between mutual funds and public firms. In contrast to their work, this research investigates the causal effects between the hiring of fund managers based on alumni connections and subsequent fund performance. Notably, while such networks might lead to lower returns, they seem to enhance fund inflows, particularly when the returns are negative in the year of manager's hiring.

Second, this paper adds to the literature on mutual funds' decision-making. Extensive research has addressed how mutual fund take risks, attract inflows, and the incentives behind these strategies (Chevalier and Ellison, 1997; Sirri and Tufano, 1998; Huang et al., 2007; Brown et al., 1996; Goetzmann et al., 2007). Specific to managerial strategies, literature has emphasized the process of fund incubation (Arteaga et al., 1998; Edelen, 1999) and termination (Brown et al., 1995; Lunde et al., 1999). Managerial decisions, particularly around promotion and demotions, have also been explored (Chevalier and Ellison, 1999; Hu et al., 2000; Baks, 2002; Khorana, 1996). This paper uniquely pinpoints the mutual fund industry's inclination to hire managers based on alumni connections,

aiming to gauge the efficiency of such hiring practices and the underlying mechanism of these alumni ties.

Third, this work closely aligns with literature on talent acquisition disparities in mutual fund industry. A salient topic in the literature concerns the gender bias within the mutual fund sector. While numerous studies show no significant performance differential between male and female fund managers, women are still significantly underrepresented (Atkinson et al., 2003; Barber and Odean, 2001). Furthermore, there’s evidence of racial barriers, with minority groups facing challenges entering the mutual fund industry, despite some improvements in diversity over time (Dewald et al., 2022; Adams and Mansi, 2023). Transitioning from these disparities to a focus on educational background, this paper unveils a preference among mutual funds for hiring managers from the same university, suggesting that a homophily bias might be influencing hiring decisions in the industry.

3 Data and Methodology

3.1 Data

3.1.1 Data Source

Our primary dataset integrates information from three pivotal sources. Initially, we extracted comprehensive mutual fund manager details from MorningStar, encompassing attributes such as the ticker, the manager’s name, fund family, gender, tenure duration, brief descriptions, and crucially, their educational background - spanning Bachelor’s to PhD degrees. Figure 1 displays the “People” section for the fund “JPMorgan Equity Income A”, visualizing manager timelines that delineate the tenure of both current and former managers, with color-coded bars representing genders. Owing to the graphic representation of the tenure, durations were approximated to the nearest year. In contract,

Figure 2 presents a detailed profile of a specific manager when selected, highlighting prior roles, educational credentials, and reported gender. Subsequently, we integrated data from the CRSP Survivor-Bias Free US Mutual Fund Database, offering a wealth of data on monthly fund returns, total net assets, diverse fund classifications, investment motives, and other salient fund characteristics, spanning a period from 2003 (coinciding with MorningStar’s records of manager’s information) to 2023. Complementing these sources, the Kenneth French website equipped us with essential market return data.

Our study focuses on actively-managed domestic equity mutual funds. We have excluded categories such as bond, money market, international, balanced, and index funds through various filters, including S&P objective, ICDI objectives, Wiesenberger Fund Type Code, and specific fund names ¹. Furthermore, due to time constraints, we limited our data collection to funds managing assets exceeding \$1 billion in May 2023.

MorningStar’s expansive and detailed coverage of mutual fund manager information has made it a favored source in academic research. For instance, for active managers, MorningStar denotes the precise inception date of a fund manager’s portfolio oversight. However, for past managers, a visual bar representation is employed, from which we deduce the start and end years of their tenure. Consequently, all metrics concerning duration are rounded to the nearest year.

MorningStar’s database on educational background details is inclusive but occasionally lacks precision. There are instances of data errors from missing information, duplicate entries, to typographical errors. An example is its broad classification of graduates from large public universities, such as Mr. Peter M. Kwiatkowski, a fund manager of

¹Firstly, we select mutual funds with the following S&P objectives: AGG, EMV, FIN, GLD, GMC, GRI, GRO, HLT, ING, NTR, RLE, SCG, SEC, TEC, UTL. If a fund does not have any of the above objective code, we select funds with the following ICDI objectives: EIEL, G, LCCE, LCGE, LCVE, MCCE, MCGE, MCVE, MLCE, MLGE, MLVE, SCCE, SCGE, SCVE, CA, EI, G, GI, MC, MR, SG, FS, H, ID, NR, S, SP, TK, TL, UT. If a fund has neither the S&P objective code nor the ICDI objective code, we select funds with the following Wiesenberger Fund Type Code: G, ENR, FIN, GCI, GPM, HLT, IEQ, LTG, MCG, SCG, TCH, UTL. We further exclude index funds and target date funds based on their names.

“AAM/Bahl & Gaynor Income Growth A” since 2019, as a graduate of “California State University” without specifying from which of its 23 campuses he graduated from. Given the necessity of precise alumni connections in our study, we manually verified the educational backgrounds of fund managers from such large institutions using LinkedIn and the official websites of their respective firms.²

Our exploration of the causal effect of alumni managers acquiring and subsequent performance required sourcing monthly return and total net asset data from the CRSP Mutual Fund Database. In the absence of exact monthly hiring timestamps, we assume all hirings take place at the year’s end in December. To ensure robustness, we also considered potential hiring dates at end of June. We evaluated each fund manager’s total asset management prior to their appointment using the monthly total net asset data. Monthly returns informed us about post-hiring performance. Additionally, using both monthly asset and return data, we assess the flow-return sensitivity for each fund post-manager hiring. The details of this process will be expanded upon in the methodology section.

3.1.2 Empirical Strategy

In this section, we focus our analysis on the concept of an “hiring event”. This term denotes the matching of a fund with a mutual fund manager at the point the manager’s name first emerges within the managerial team of the fund. Such an event spans both “active” process (e.g., hirings) and “passive” integration, as seen in instances of fund mergers or acceptances following a fund merge.

Each such event constitutes a singular observation in our analytical framework. For example, should a manager be hired by several mutual funds within a year, every matching

²We used LinkedIn and official websites of mutual fund manager’s respective firms to improve the accuracy for managers graduated from the following institutions: California State University, City University of New York, State University of New York, Indiana University, University of Colorado, University of California, University of Illinois, University of Texas, University of Wisconsin.

is counted as an independent hiring. As an illustrative example, when one manager is hired by five distinct mutual funds in year t , we document this as five separate hiring instances. A parallel approach applies if one fund integrates multiple managers in a single year, each is recognized independently.

However, our analysis framework omits situations where a fund maintains a manager beyond their acceptance year. And due to the difficulties of determining in 2003 - whether a fund hired a manager that year or beforehand - we have excluded all such entries for 2003. Consequently, our dataset shows 1,809 distinct mutual funds, resulting in 14,345 manager hiring events spanning 2004 to 2023.

Table 1 presents summary statistics for fund-manager pairings from 2004 to 2023. Panel A delineates fundamental metrics surrounding fund managers: tenure, the number of funds that a given manager is managing in given year, survival indicator variables. Within our sample, data arises from 97 distinct mutual fund families, capturing 2,501 individual mutual funds and 1,809 unique fund managers. The Asset Under Management (AUM) in our sample ranges from \$0.1 million to \$256,318.8 million, averaging at \$12,366.64 million. The average managerial tenure is 6.52 years, with the mean number of funds managed by a particular manager at 12.66. Over a span of 2 to 5 years, the retention rate for managers across from 75.39% to 48.5%.

Panel B provides statistics on various manager attributes, such as gender distribution and educational backgrounds. The data reveals 87.39% male fund managers, opposed to 12.61% female fund managers. An overwhelming 97% of these managers hold a Bachelor's degree. MBA degree holders account for 59.19%, while only 5.09% possess a PhD or equivalent degree. Regarding to alumni affiliations, 20.66% of managers share academic ties with at least one peer manager in the same fund, and 57.38% within the broader fund family.

Panel C includes the top-tier academic institutions cultivating these managers. Among

the 1,809 managers, institutions like Harvard University, University of Pennsylvania, University of Chicago, Columbia University, and New York University contribute to over 5% of the managerial talent pool.

Figure 3, a time series plot, outlines the annual trend in fund manager hiring from 2004 to 2023. The highlighted periods encapsulate the 2008-2009 global financial crisis and the 2020 COVID-19 pandemic - epochs recognized as disrupts for the financial sector and its talent sourcing. Observably, we see a slight decline in hiring in 2008, but the pandemic of 2020 precipitated a more pronounced downturn in managerial hiring.

3.1.3 Sample Construction

Within the MorningStar database, we are only able to observe the fund-manager hiring pairs in which mutual funds made actual decisions to incorporate managers. However, the central question is whether an alumni connection between the current fund managers and newly hired fund manager increases the latter’s likelihood of hired by the fund. Thus, we provide a structured analysis of this primary research question, suggesting that pre-existing alumni relationships enhance the probability of a fund-manager matching.

To evaluate the influence of these alumni ties on managerial hiring process, it is important to consider for both the actual hiring pairs and the hypothetical or ”counterfactual” pairs. The latter represent manager who were likely considered during the recruitment process but were ultimately not hired by funds. This perspective broadens our frame of analysis to include potential matches between a fund and a manager. Consequently, our study is not biased to just the realized hirings (where a fund successfully matches a manager in a given year). It also extends to include scenarios where a fund might have considered on acquiring a manager but eventually refrained.

To create a plausible set of these potential fund-manager parings, I construct a “candidate pool” for each year, t . This pool comprises fund managers actively exploring new

opportunities within the industry during that year. Specifically, the “candidate pool” includes all managers who undertook at least one new role in year t . That is, we identify counterfactual candidates: managers that funds probably considered for acquiring, but ultimately did not take the action.

With the assumption that mutual funds evaluate potential managers based on comparable credentials and past achievements, I categorize managers into groups, segmented by Asset Under Management (AUM) quintiles and quintiles of the number of funds managed. For instance, if a fund hired Ms. Smith in year t , other managers within the same group of AUM and past management experience as Ms. Smith would be accounted for plausible candidates for that fund in our framework.

With this methodology, our sample considered both the actually hired managers and their conjectural counterparts, resulting in a dataset of 1,914,368 fund-manager combinations.

Table 2 reports the summary statistics of key variables of the full matching sample. The average hiring is 0.0075, meaning that on average each mutual fund manager hiring correlates with approximately 133 counterfactual candidates. The table also shows the percentage of alumni affiliations within the fund, within the family, and the percentage of connections established through undergraduate, masters, MBA and doctoral programs.

3.1.4 Empirical Specification

To test our main hypothesis, we employ an Ordinary Least Square regression model to test the likelihood of a manager’s hiring by a mutual fund. This evaluation is performed on the comprehensive sample, including both actual and counterfactual manager hiring. The equation can be denoted as:

$$Hire_{f,m,t} = \alpha_t + \gamma_f + \eta_m + \beta_1 \times Alumni_{f,m} + \epsilon_{f,m,t} \quad (1)$$

where $Hire_{f,m,t}$ is a binary variable indicating whether the mutual fund f hires manager m in year t ; $Alumni_{f,m}$ is a indicator variable set to 1 if fund manager m shares alumni ties with at least one incumbent manager in fund f ; α_t is year fixed effect; γ_f is fund fixed effect; and η_m is fund manager fixed effect. The standard error is clustered by funds. This specification allow us to test the hypothesis that when a fund manager has alumni connection with the current fund manager, she stands a better chance to be hired by the fund.

Table 3 reports the results from our OLS regressions. Column 1 to column 3 indicate that mutual funds are 0.9% more likely to hire alumni fund manager who graduated from the same university as current fund managers employed at that fund when we include fund, manager and year fixed effects. Column 4 shows that mutual funds are 1.04% more likely to hire alumni fund manager if we loose the alumni connections from intra-fund to intra-family. Column 5 examines the disparity in alumni effects across academic degrees of the newly hired managers. The results show that fund managers who obtain alumni ties through undergraduate and MBA programs are significantly affected, which implies that when making hiring decisions, mutual funds value Bachelor's and MBA degrees more.

For additional tests, we also examine the alumni connection effects if we split the sample based on varying team sizes of the funds, early vs. later years, and recession vs. normal years. We will show details in later sections.

3.2 Heterogeneous Effects

To understand the effects of alumni ties on fund manager hiring under various circumstances, we test the heterogeneous effects under different fund characteristics, years and macroeconomic conditions.

3.2.1 Fund Characteristic – Large Fund vs. Small Fund

We firstly calculated the manager’s previous year’s Total Net Asset (TNA). We then created an indicator variable set to 1 for funds with a TNA greater than the median, and 0 otherwise. This allows us to classify funds into “large” and “small” categories. Table 4 demonstrates the alumni connection effects for both. Large funds had 1.07% (19.5% more relative to the baseline) increased likelihood of hiring managers with alumni connections. On the other hand, small funds displayed a 0.85% (5.19% less relative to the baseline) reduced probability. A significant trend emerged for large funds, their preference of MBA alumni ties increased 24.5% relative to the baseline finding, while small funds’ preference of MBA alumni ties decreased 10.2% relative to the baseline. The results indicate that fund alumni ties established in MBA programs play a more influential role in large funds.

3.2.2 Early Years vs. Later Years

We categorized hirings as “early” (before 2014) or “later” (from 2014 onwards). Table 5 reflects a growing emphasis on alumni ties in the recent years. The data indicates an increase of 21.0% for preference of in-fund alumni connections and 22.3% for in-family ties. The value placed on ties formed during undergraduate and MBA programs also surged, by 55.9% and 56.6% respectively, in the latter years.

3.2.3 Market Characteristic – Recession Time vs. Normal Time

How do talent hiring mechanism changes in responding different macroeconomic environment? For economic condition analysis, we split our data based on the U.S. unemployment rate. We designated an indicator variable, *HighUnemployment*, set to 1 for years with unemployment rates above 6% and 0 otherwise. Table 6 reveals that during high unemployment years, the attraction of alumni networks strengthens. In detail, We find

that the family network effects influence more in high unemployment years, leading mutual funds a 1.28% (22.8% more relative to baseline) more likely to hire fund managers who have graduated from the same university as current managers employed at given fund family. Comparatively, we observe that in high unemployment years, funds prefer family-alumni connections by 34.78% more than they do in normal year. In addition, newly hired managers received more assistance by fund alumni ties through undergraduate and MBA programs in high unemployment years than that in normal unemployment years.

Furthermore, we used a different metric, the National Bureau of Economic Research (NBER) Recession index, to differentiate normal time and recession time, which is more prevalent in financial economic research. We split the sample and label *NBER – Recession* equals to 1 for 2008, 2009 and 2020, otherwise label *NBER – Recession* equals to 0. From Table 7, we find that the fund alumni ties, family alumni ties, and fund alumni through PhD programs are more pronounced during recession years. More specifically, funds are 1.06% more likely to hire managers who have fund alumni connections (14.7% more relative to normal years), 1.38% more likely to hire managers who have family alumni connections (33.7% more relative to normal years), and 1.78% more likely to hire managers who have fund connections through their PhD programs.

Regardless of the measurement employed, alumni connections consistently emerge as a important factor in mutual fund talent acquisition, especially during economically strenuous times.

4 Post Acquisition Performance Evaluation

4.1 Sample Construction

We are interested in understanding how alumni ties influence a fund manager’s subsequent performance in areas like returns, duration of service, survival likelihood, and ability to attract investment flows. Past studies suggest that manager returns often correlate with their experience, educational background from elite universities, gender, and prevailing macroeconomic conditions. Another observation from the literature is that managers with better past performance tend to serve longer, have higher survival rates, and attract more funds.

To navigate these studies, we assess the performance differences between managers with and without alumni ties using the nearest neighbor propensity score matching approach (as detailed in Rosenbaum and Rubin (1983), Smith and Todd (2005)). Propensity score matching method is gaining attention for its efficiency in forming comparison groups in non-experimental research. Our aim is to pair managers with alumni ties (at hiring) with those without, ensuring no observable difference in their characteristics. More importantly, this method can accommodate a large number of matching variables as well.

Our first step involves determining the likelihood of a manager, based on their profile, having alumni ties with current managers at the fund. For this, we employ a probit regression to understand this likelihood against manager-level factors. Specifically, we include the fund family that the manager was associated with before being hired, a dummy variable indicating whether the manager graduated from a top-tier university, the manager’s gender, the manager’s AUM and number of funds they managed the year before hiring.

³ To ensure accurate matching, we factor in the year during our propensity score esti-

³We identify the variable *TopUniversity* as a dummy variable if the manager obtains at least one of her degrees from the following universities: Princeton University, Brown University, Columbia University, Harvard University, University of Pennsylvania, Cornell University, Yale University, Dartmouth College, University of Chicago, Massachusetts Institute of Technology, Stanford University, and University of

mation, ensuring that both groups of managers operate under identical macroeconomic conditions.

Using data for 2,964 fund-manager hiring with alumni ties and 11,381 without, we first estimate the propensity scores via the probit regression. We then pair each manager with alumni ties to their closest match from the group without ties, based on the propensity score. In our further tests, we broaden our matching criteria, pairing with 5 nearest matches. Our findings remain consistent, even with this relaxed approach.

4.2 Subsequent Performance

4.2.1 Returns

Table 8’s initial section examines the effect of fund-manager hirings related to alumni ties on their subsequent one and two-year returns. A fascinating discovery is that managers with alumni ties exhibit an average one-year raw return of 1.75%, whereas their counterparts without such ties boast a 3.1% return. This translates to a underperformance of 1.4% for the former group in the year post-hiring. Moreover, this performance gap widens to 1.6% if we evaluate two-year returns after hiring.

Such results hint that discrepancies in future returns due to alumni ties might stem from challenges the newly hired manager faces when collaborating with the existing team post-hiring.

4.2.2 Tenure and Survival Rate

The following section of Table 8 shows how these alumni-related hirings influence managers’ tenure and their survival rate afterwards. On the tenure front, managers with alumni ties average a service length of 6.58 years, while those without these ties serve

California, Berkeley.

for about 7.24 years. Given the noted under-performance of the alumni-tied group in subsequent years, this reduced tenure - by about 0.67 years - might be attributed to that under-performance.

We also explore survival rates across different time spans (2, 3, 5, and 10 years) post-hiring for both groups. Managers with alumni ties display a 78.0% survival probability over 2 years, a little lower than the 80.5% for those without ties. A similar pattern holds for the 3-year and 10-year periods, although the 5-year survival rate does not display a significant difference. Notably, these findings underscore that observed differences in manager performance are not merely because of any evident variations in their individual attributes.

4.2.3 Flow-Return Sensitivity

Mutual fund companies generate revenue from a fixed percentage of their Asset Under Management (AUM). This establishes a strong incentive for them to attract substantial inflows. Previous literature has recognized a pronounced positive relationship between new investment inflows into a fund and its past performance, terming it "return chasing". This incentive essentially drives mutual funds to deliver better performance to their investors.

In this section, what we examine the flow-performance relationship to understand the potential incentives for funds that hire managers based on alumni connections, especially when past fund performance has been underperforming. We aim to answer two crucial questions: Does acquiring a manager with alumni ties bolster fund inflow? Can such in hiring help sustain the existing AUM, preventing outflows, when the fund has underperformed in the year of manager hiring? To address these hypothesis, we employ two regression models on our sample sourced from the Propensity Score Matching Algorithm:

$$Flow_{m,t+1} = \alpha_{f,m,t} + \beta_1 \times Return_{m,t} + \beta_2 \times Return_{m,t} \times Alumni_{f,m} + \epsilon_{f,m,t}^1 \quad (2)$$

$$Flow_{m,t+2} = \gamma_{f,m,t} + \beta_1 \times Return_{m,t} + \beta_2 \times Return_{m,t+1} + \beta_3 \times Return_{m,t} \times Alumni_{f,m} + \epsilon_{f,m,t}^2 \quad (3)$$

where in the equation 2, $Flow_{m,t+1}$ is the weighted average percentage of fund flow of the manager m for the year $t + 1$, $Return_{m,t}$ represents the manager m 's weighted average return for all her new funds in year t . $Alumni_{f,m}$ is a dummy variable, indicating if the given manager m has alumni connections with current managers of fund f . Similar to equation 2, in equation 3, $Flow_{m,t+2}$ is the weighted average percentage of fund flow of the manager m for the year $t + 2$, $Return_{m,t+1}$ is the weighted average return for manager m in year $t + 1$.

Table 9 reports the flow-return sensitivity using our sample. A discernible "return chasing" pattern emerges. Specifically, for our complete sample, a 1% increase in current year manager's return bolster a 22.7% surge in the following year's inflow. When segmented the sample based on positive and negative current-year manager returns, a 1% drop in returns results in a 3.67% fund outflow. Intriguingly, the presence of an alumni connection seems to reverse this trend. Managers hired by alumni connected funds experience a 27% outflow for a 1% positive return, but when the current year manager's returns is negative, these funds surprisingly benefit from a 3.89% inflow, compared to the 3.67% outflow observed in other funds.

Our findings in this section suggest that investors' perceptions might significantly vary based on the internal governance of the fund, especially when performance varies. A positive return may make investors worry about potential agency issues arising from alumni hires, prompting redemption. Conversely, during underperformance, these same alumni ties might be perceived as a potential future performance booster, encouraging

investment inflow.

Given our dataset’s annual granularity, a more detailed, perhaps monthly, dataset might enhance our understanding of these dynamics.

5 Robustness Check

5.1 Talent Acquisition: Logit Model and Conditional Choice Model

Table 3 lends credence to our primary result that a prior alumni connection increases the chance of a fund manager being hired by a fund. To further substantiate this claim, we employed both the Logit Model and the Conditional Choice Model as robustness checks.

In Table 10, we present the results using the logit model, detailing how alumni ties influence mutual fund talent acquisition. The dependent variable *Hire*, denotes whether a particular fund hires a manager. The independent variables remain consistent with equation 1. The regression coefficient for ‘*FundAlumni*’ is positive and highly significant at the 1% level. This indicates that an existing alumni tie enhances a manager’s likelihood of hiring by the fund by approximately 0.82%. Such findings support our primary hypothesis, underlining the alumni connection’s role in the manager hiring process. Further analysis, as seen in the second and third columns, echo these findings and the results from our baseline analysis. We discern that alumni connections established either through undergraduate or MBA programs retain their significance at the 1% level.

Inspired by Hacamo and Kleiner (2022), we incorporated a conditional choice model to ascertain whether a manager, ‘*m*’, with alumni connections to a fund, ‘*f*’, is more inclined to associate with the fund. Table 11 reports our findings in the form of odds ratios. The implications of having an alumni tie within a fund on the hiring likelihood stand at 0.92%. Furthermore, the influence of a family alumni connection on the hiring

change is approximately 0.98%. When examining the educational programs fostering these alumni connections, it emerges that ties established through undergraduate, MBA, and PhD levels play a pivotal role in facilitating fund hirings.

5.2 Talent Acquisition: Candidate Pool Shrinkage

In section 3.1.3, our focus lies in constructing a candidate pool for every managerial hiring instance. We implemented additional filters to condense our sample size. This was done to adhere to our "candidate pool" hypothesis: managers possessing similar qualifications and past experience are perceived as potential candidates by mutual funds searching for new managerial talent. To further streamline this selection, we clustered managers into distinct groups based on the Asset Under Management (AUM) quintiles and the quintiles of amount of funds managed one year before. These filters grant us a more concise and reasonable sample to investigate the implications of alumni connections on talent acquisition. Table 12 shows the OLS regression results from our shrunked sample. Although the sample size is smaller, the coefficient's magnitude has increased, with the directional trend remaining the same as our baseline results. Specifically, the tendency of funds to incline more preference towards managers with established alumni ties remain discernible, especially if these connections were cultivated during undergraduate or MBA programs.

However, despite our rigorous filtering process, further refining the sample to mirror real-world dynamics is plausible. Consider the geographical factor: a fund headquartered in Boston, Massachusetts might be inclined to hire a new manager from the U.S. East Coast, but a manager based in California or Texas might be less inclined to reallocate to Boston. If we had access to geographical data regarding fund managers' residences, it could potentially enhance the precision of our conclusions.

5.3 Performance Evaluation: Propensity Score Matching Control Group

To ensure the robustness of our findings on the effect of alumni hires on manager performance, we modified the propensity score matching algorithm by changing the number of matched neighbors from 1 to 5. This modification is crucial in ascertaining if the direction and magnitude of our observed effects are consistent when considering a broader set of compares, rather than just the closest match. It also helps mitigate the risk of any potential bias introduced by an overly restrictive matching process.

Upon adjusting the number of neighbors in the propensity score matching, our findings showed in Table 13 and 14 indicate a consistent negative relation between alumni hires and subsequent manager performance, though the magnitude of the effect increased slightly. Moreover, the one- and two-year three factor factor turn to slightly positive, meaning that alumni hires bring better subsequent manager risk adjusted return. Regarding to flow-return sensitivity, our findings are consistent with the previous findings when we have one-on-one matching of alumni hires and non-alumni hires.

This additional robustness check reinforces our primary conclusion, suggesting that the effects we observed are not solely artifacts of the specific matching parameters initially employed but are consistent even when the criteria are broadened.

6 Conclusion

Networks play an important role in the dynamics of the mutual fund industry. Mutual fund managers not only actively expand these networks but also leverage them to their advantage. Although there is a board agreement on the instrumental role of these networks in talent acquisition within the mutual fund, studying the intricacies of how these networks influence the talent acquisition process can provide invaluable insights for funds, investors, and regulatory authorities.

This research seeks to explain i) The impact of alumni networks on mutual fund talent acquisition, and ii) The performance implications of acquiring alumni managers versus those without such connections. By scrupulously constructing a candidate pool for each year's managerial hirings, we demonstrate that managers endowed with alumni ties to current fund managers stand a greater chance of being selected by the fund. This inclination is more pronounced in larger funds, and becomes especially evident in recent years and during recession periods.

However, there's a flip side. Despite the evident proclivity for acquiring alumni managers, their performance often leaves room for improvement. Managers with alumni ties tend to generate lower raw returns, have shorter tenures, and exhibit a lower survival rate compared to their counterparts without such connections. This sub-optimal performance could be indicative of mutual funds' strong bias towards alumni, which might incorporate a less optimal amount of capital towards them. Interestingly, these alumni-connected managers exhibit a buffer for preserving capital during times when they have a negative return in their hiring year, a period when other managers typically witness fund outflows.

Looking ahead, there is a vast expansion opportunities ripe for exploration. One could extend to the talent acquisition strategies of institutional investors by dissecting the implications of alumni connections between recently on-boarded managers and a fund's board members. Furthermore, an in-depth analysis of the mechanisms governing

promotions and demotions within fund families, along with the shifts incentives pre and post transitions, could provide enriching insights into the operational intricacies of the mutual fund industry.

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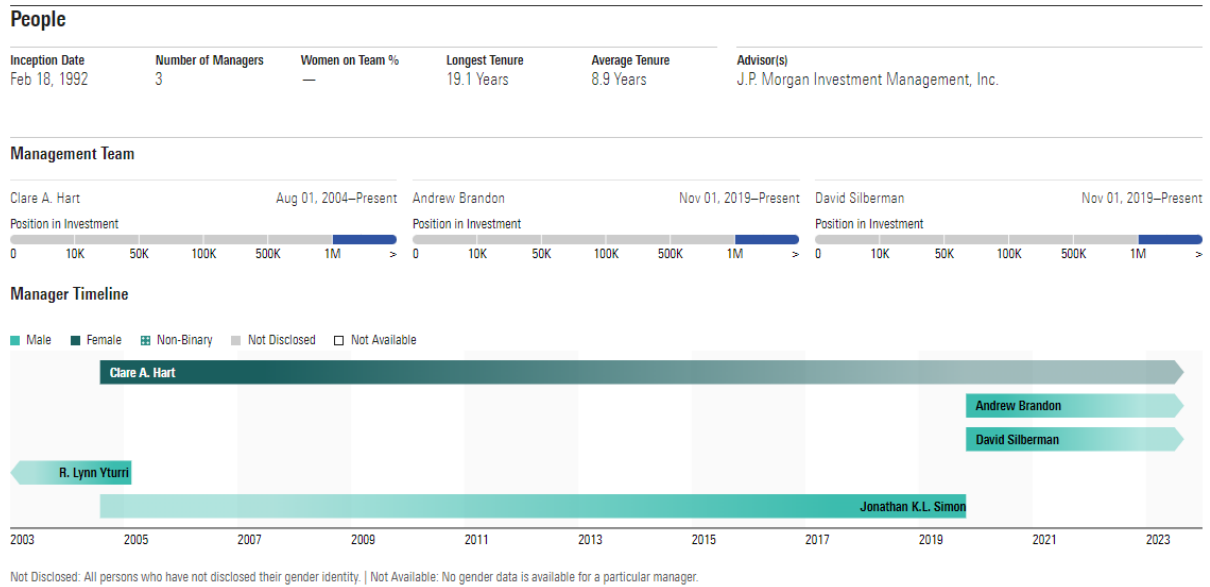


Figure 1: An Example of MorningStar Fund Information Page

Table 1: Summary Statistics

	Panel A: Fund Managers Characteristics				
	Mean	Std.	Min.	Median	Max.
AUM (million \$)	12,366.64	29,154.76	0.00	2,138.80	256,318.80
Tenure (year)	6.52	4.88	0.00	5.00	19.00
Number of Fund Managing	12.66	10.43	1.00	9.00	76.00
Survived After 2 Years	0.75	0.43	0.00	1.00	1.00
Survived After 3 Years	0.66	0.47	0.00	1.00	1.00
Survived After 5 Years	0.49	0.50	0.00	0.00	1.00
Number of Unique Families	97				
Number of Unique Funds	2,501				
Number of Unique Managers	1,809				
Manager Hirings Obs.	14,345				

Panel B: Percentages for Manager Characteristics

	Percentage
Fund Alumni	20.66%
Family Alumni	57.38%
Fund Alumni through Bachelor	5.99%
Fund Alumni through Master	0.27%
Fund Alumni through MBA	11.78%
Fund Alumni through PhD	0.03%
Male Fund Manager	87.39%
Female Fund Manager	12.61%
Holds Bachelor's Degree	97.08%
Holds Master's Degree	15.87%
Holds MBA Degree	59.19%
Holds PhD Degree	5.09%

Panel C: Top 10 University Alumni

Rank	University	Number of Manager
1	Harvard University	162
2	University of Pennsylvania	156
3	University of Chicago	142
4	Columbia University	116
5	New York University	91
6	Stanford University	81
7	University of Wisconsin, Madison	79
8	Dartmouth College	54
9	University of Michigan, Ann Arbor	52
10	University of Virginia	50

Notes: This table shows summary statistics for mutual fund managers appearing in the fund-manager acquisition dataset. Panel A shows the statistics of manager characteristics. Panel B presents the percentage of fund managers with certain characteristics in the whole data. Panel C shows the top ten universities that foster the most managers and their numbers in the dataset.

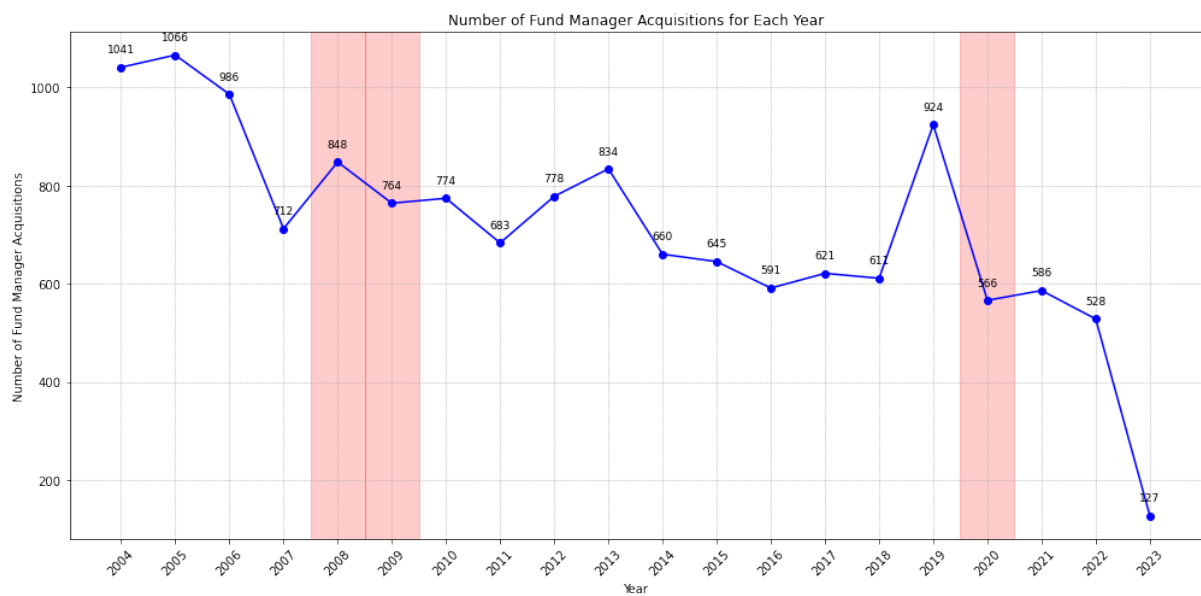


Figure 3: Number of Fund-Manager Acquisition per year from 2004 to 2023

Table 2: Summary Statistics

Panel A: Full Matching Sample				
	Mean	Std.	Min.	Max.
Hire (%)	0.75	8.62	0.00	100.00
FundAlumni (%)	10.65	30.85	0.00	100.00
FamilyAlumni (%)	37.07	48.30	0.00	100.00
FundAlumni×Bachelor (%)	4.34	20.37	0.00	100.00
FundAlumni×Master (%)	0.92	9.56	0.00	100.00
FundAlumni×MBA (%)	6.80	25.18	0.00	100.00
FundAlumni×PhD (%)	0.45	6.70	0.00	100.00
Observations	1,914,368			
Panel B: Shrunked Matching Sample				
Hire (%)	1.15	10.64	0.00	100.00
FundAlumni (%)	10.01	30.02	0.00	100.00
FamilyAlumni (%)	35.44	47.83	0.00	100.00
FundAlumni×Bachelor (%)	3.93	19.43	0.00	100.00
FundAlumni×Master (%)	0.88	9.36	0.00	100.00
FundAlumni×MBA (%)	6.43	24.53	0.00	100.00
FundAlumni×PhD (%)	0.43	6.55	0.00	100.00
Observations	1,252,711			

Notes: This table shows summary statistics for full matching sample and shrunked matching sample, including actual manager acquisitions and potential but ultimately unhappened acquisitions.

Table 3: Alumni Network Effects on Mutual Fund Talent Acquisition

	(1)	(2)	(3)	(4)	(5)
			Hire (%)		
FundAlumni	0.850*** (0.043)	0.831*** (0.043)	0.892*** (0.041)		
FamilyAlumni				1.038*** (0.024)	
FundAlumni×Bachelor					0.628*** (0.048)
FundAlumni×Master					-0.154* (0.083)
FundAlumni×MBA					0.972*** (0.052)
FundAlumni×PhD					0.075 (0.117)
Fund FE	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Manager FE	No	No	Yes	Yes	Yes
Observations	1,914,368	1,914,368	1,914,368	1,914,368	1,914,368
R-Squared	.000961	.00135	.00533	.00649	.00544

Notes: This table estimates the effects of alumni networks on fund manager hiring by

$$Hire_{f,m,t} = \alpha_t + \gamma_f + \eta_m + FundAlumni_{f,m} + \epsilon_{f,m,t} \quad (4)$$

fund, manager and year fixed effects are included. The dependent variable in all specifications is an indicator variable denoting that the fund manager m is acquired by mutual fund f in year t . FundAlumni is an indicator variable denoting that the candidate went to the same university as an individual who was a fund manager in the fund at the time of hiring. FamilyAlumni is an indicator variable denoting that the candidate went to the same university as an individual who was a fund manager in the family at the time of hiring. FundAlumni×Bachelor is an indicator variable denoting that the candidate obtained alumni ties through undergraduate programs at the time of hiring. FundAlumni×Master is an indicator variable denoting that the candidate obtained alumni ties through master programs, but excluding MBA programs at the time of hiring. FundAlumni×MBA is an indicator variable denoting that the candidate obtained alumni ties through MBA programs at the time of hiring. FundAlumni×PhD is an indicator variable denoting that the candidate obtained alumni ties through PhD programs at the time of hiring.

Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 4: Alumni Network Effects on Mutual Fund Talent Acquisition for Different Team Size

	(1)	(2)	(3)	(4)	(5)	(6)
	Large Funds Hire (%)		Small Funds Hire (%)			
FundAlumni	1.065*** (0.068)			0.848*** (0.050)		
FamilyAlumni		1.087*** (0.041)			1.070*** (0.027)	
FundAlumni×Bachelor			0.629*** (0.078)			0.609*** (0.063)
FundAlumni×Master			-0.127 (0.132)			-0.103 (0.107)
FundAlumni×MBA			1.212*** (0.089)			0.882*** (0.063)
FundAlumni×PhD			0.104 (0.210)			0.133 (0.149)
Fund FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Manager FE	Y	Y	Y	Y	Y	Y
Number of obs.	631,326	631,326	631,326	1,283,042	1,283,042	1,283,042
R-squared	0.010	0.011	0.011	0.006	0.007	0.006

Notes: This table shows the OLS estimation of alumni networks effects on mutual fund talent acquisition when we split the full matching sample to large funds and small funds.

Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 5: Alumni Network Effects on Mutual Fund Talent Acquisition for Different Time Periods

	(1)	(2)	(3)	(4)	(5)	(6)
	Early Years Hire (%)		Later Years Hire (%)			
FundAlumni	0.872*** (0.044)			1.079*** (0.066)		
FamilyAlumni		0.993*** (0.026)			1.269*** (0.043)	
FundAlumni×Bachelor			0.528*** (0.056)			0.823*** (0.080)
FundAlumni×Master			-0.030 (0.100)			-0.168 (0.144)
FundAlumni×MBA			0.847*** (0.052)			1.326*** (0.087)
FundAlumni×PhD			0.688*** (0.167)			-0.803*** (0.179)
Fund FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Manager FE	Y	Y	Y	Y	Y	Y
Number of obs.	1,248,791	1,248,791	1,248,791	665,577	665,577	665,577
R-squared	0.004	0.005	0.004	0.007	0.009	0.008

This table shows the OLS estimation of alumni networks effects on mutual fund talent acquisition when we split the full matching sample to early years and later years based on the year cutoff of 2013.

Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 6: Alumni Network Effects on Mutual Fund Talent Acquisition for High Unemployment Rate Years and Normal Unemployment Rate Years

	(1)	(2)	(3)	(4)	(5)	(6)
	Above 6% Unemployment Hire (%)		Below 6% Unemployment Hire (%)		Unemployment Hire (%)	
FundAlumni	0.978*** (0.050)			0.927*** (0.051)		
FamilyAlumni		1.275*** (0.035)			0.946*** (0.028)	
FundAlumni×Bachelor			0.836*** (0.073)			0.459*** (0.059)
FundAlumni×Master			-0.409*** (0.112)			0.213* (0.109)
FundAlumni×MBA			0.915*** (0.059)			1.127*** (0.070)
FundAlumni×PhD			0.110 (0.152)			0.175 (0.209)
Fund FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Manager FE	Y	Y	Y	Y	Y	Y
Number of obs.	832,475	832,475	832,475	1,081,893	1,081,893	1,081,893
R-squared	0.005	0.007	0.005	0.006	0.007	0.006

This table shows the OLS estimation of alumni networks effects on mutual fund talent acquisition when we split the full matching sample to recession years when the U.S. Unemployment rate is above 6%, and normal years elsewhere. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 7: Alumni Network Effects on Mutual Fund Talent Acquisition for Recession Time and Normal Time

	(1)	(2)	(3)	(4)	(5)	(6)
	Recession Years		Non-recession Years			
	Hire (%)		Hire (%)			
FundAlumni	1.061*** (0.090)		0.925*** (0.045)			
FamilyAlumni		1.382*** (0.059)			1.034*** (0.025)	
FundAlumni×Bachelor			0.669*** (0.132)			0.629*** (0.053)
FundAlumni×Master			-0.163 (0.207)			-0.089 (0.080)
FundAlumni×MBA			0.916*** (0.102)			1.044*** (0.060)
FundAlumni×PhD			1.776*** (0.401)			-0.233* (0.124)
Fund FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Manager FE	Y	Y	Y	Y	Y	Y
Number of obs.	269,225	269,225	269,225	1,645,143	1,645,143	1,645,143
R-squared	0.007	0.009	0.007	0.005	0.006	0.005

This table shows the OLS estimation of alumni networks effects on mutual fund talent acquisition when we split the full matching sample to recession years and normal years according to NBER Recession Index.

Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 8: Propensity Score Matching Estimators

	Nearest neighbor is 1			
	No. of obs.	Mean	Diff. (Alumni - NonAlumni)	P-value of diff.
<i>Return</i> _{<i>t</i>+1} (Alumni)	4,202	0.017	-0.014***	0.000
<i>Return</i> _{<i>t</i>+1} (NonAlumni)		0.031	(0.002)	
<i>Alpha</i> _{<i>t</i>+1} (Alumni)	4,202	0.001	0.001	0.131
<i>Alpha</i> _{<i>t</i>+1} (NonAlumni)		-0.000	(0.001)	
<i>Return</i> _{<i>t</i>+1} + <i>Return</i> _{<i>t</i>+2} (Alumni)	4,024	0.018	-0.016***	0.000
<i>Return</i> _{<i>t</i>+1} + <i>Return</i> _{<i>t</i>+2} (NonAlumni)		0.034	(0.002)	
<i>Alpha</i> _{<i>t</i>+1} + <i>Alpha</i> _{<i>t</i>+2} (Alumni)	4,024	0.002	0.002**	0.030
<i>Alpha</i> _{<i>t</i>+1} + <i>Alpha</i> _{<i>t</i>+2} (NonAlumni)		0.001	(0.001)	
<i>Flow</i> _{<i>t</i>+1} (Alumni)	4,202	0.028	-0.090***	0.000
<i>Flow</i> _{<i>t</i>+1} (NonAlumni)		0.119	(0.244)	
<i>Tenure</i> (Alumni)	5,928	6.575	-0.669***	0.000
<i>Tenure</i> (NonAlumni)		7.244	(0.127)	
<i>Survive</i> _{<i>t</i>+2} (Alumni)	5,928	0.780	-0.026**	0.015
<i>Survive</i> _{<i>t</i>+2} (NonAlumni)		0.805	(0.011)	
<i>Survive</i> _{<i>t</i>+3} (Alumni)	5,928	0.695	-0.046***	0.001
<i>Survive</i> _{<i>t</i>+3} (NonAlumni)		0.741	(0.012)	
<i>Survive</i> _{<i>t</i>+5} (Alumni)	5,928	0.492	-0.007	0.59
<i>Survive</i> _{<i>t</i>+5} (NonAlumni)		0.499	(0.013)	
<i>Survive</i> _{<i>t</i>+10} (Alumni)	5,928	0.209	-0.095***	0.000
<i>Survive</i> _{<i>t</i>+10} (NonAlumni)		0.304	(0.011)	

Notes: This table estimates the average treatment effect of treated group by using the propensity score matching algorithm. In this algorithm, the treatment group includes manager acquisition observations of which the acquired manager has alumni ties with the current fund manager. The control group contains manager acquisition observations of which the acquired manager does not have alumni connections with the current fund manager. In the propensity score matching algorithm, we have one-on-one matching of an alumni-acquisition and a non-alumni acquisition. We tested the difference of one-year return (and excess return), two-year return (excess return), tenure, and survival rate after 2, 3, 5, and 10 years of acquisition. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 9: Flow Return Sensitivity

	(1)	(2)	(3)
	Full Sample	$Flow_{t+1}$ $Return_t < 0$	$Return_t > 0$
$Return_t$	0.227*** (0.050)	-0.037*** (0.009)	0.207*** (0.060)
$Return_t$ $\times FundAlumni$	-0.043*** (0.013)	0.002 (0.002)	-0.090*** (0.029)
Number of obs.	4,120	806	3,314
R-squared	0.002	0.063	0.002

Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 10: Alumni Network Effects on Mutual Fund Talent Acquisition – Logit Model

	(1)	(2)	(3)
	Hire (%)		
FundAlumni	0.820*** (0.026)		
FamilyAlumni		0.861*** (0.017)	
FundAlumni×Bachelor			0.508*** (0.032)
FundAlumni×Master			-0.200** (0.082)
FundAlumni×MBA			0.810*** (0.024)
FundAlumni×PhD			0.240** (0.098)
Number of obs.	1,914,368	1,914,368	1,914,368

Notes: This table estimates the effects of alumni networks on fund manager acquisition by using the conditional choice model. The dependent variable in all specifications is an indicator variable denoting that the fund manager m is acquired by mutual fund f in year t . FundAlumni is an indicator variable denoting that the candidate went to the same university as an individual who was a fund manager in the fund at the time of hiring. FamilyAlumni is an indicator variable denoting that the candidate went to the same university as an individual who was a fund manager in the family at the time of hiring. FundAlumni×Bachelor is an indicator variable denoting that the candidate obtained alumni ties through undergraduate programs at the time of hiring. FundAlumni×Master is an indicator variable denoting that the candidate obtained alumni ties through master programs, but excluding MBA programs at the time of hiring. FundAlumni×MBA is an indicator variable denoting that the candidate obtained alumni ties through MBA programs at the time of hiring. FundAlumni×PhD is an indicator variable denoting that the candidate obtained alumni ties through PhD programs at the time of hiring.

Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 11: Alumni Network Effects on Mutual Fund Talent Acquisition – Conditional Choice Model

	(1)	(2)	(3)
	Hire (%)		
FundAlumni	0.916*** (0.023)		
FamilyAlumni		0.985*** (0.018)	
FundAlumni×Bachelor			0.572*** (0.032)
FundAlumni×Master			-0.147* (0.082)
FundAlumni×MBA			0.897*** (0.026)
FundAlumni×PhD			0.306*** (0.098)
Number of obs.	1,914,368	1,914,368	1,914,368

Notes: This table estimates the effects of alumni networks on fund manager acquisition by using the conditional choice model. The dependent variable in all specifications is an indicator variable denoting that the fund manager m is acquired by mutual fund f in year t . FundAlumni is an indicator variable denoting that the candidate went to the same university as an individual who was a fund manager in the fund at the time of hiring. FamilyAlumni is an indicator variable denoting that the candidate went to the same university as an individual who was a fund manager in the family at the time of hiring. FundAlumni×Bachelor is an indicator variable denoting that the candidate obtained alumni ties through undergraduate programs at the time of hiring. FundAlumni×Master is an indicator variable denoting that the candidate obtained alumni ties through master programs, but excluding MBA programs at the time of hiring. FundAlumni×MBA is an indicator variable denoting that the candidate obtained alumni ties through MBA programs at the time of hiring. FundAlumni×PhD is an indicator variable denoting that the candidate obtained alumni ties through PhD programs at the time of hiring.

Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 12: Alumni Network Effects on Mutual Fund Talent Acquisition – Sample Shrinkage

	(1)	(2)	(3)	(4)	(5)
			Hire (%)		
FundAlumni	1.480*** (0.075)	1.451*** (0.075)	1.500*** (0.068)		
FamilyAlumni				1.616*** (0.037)	
FundAlumni×Bachelor					1.049*** (0.081)
FundAlumni×Master					-0.151 (0.130)
FundAlumni×MBA					1.634*** (0.085)
FundAlumni×PhD					0.203 (0.197)
Fund FE	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Manager FE	No	No	Yes	Yes	Yes
Number of obs.	1,252,701	1,252,701	1,252,701	1,252,701	1,252,701
R-squared	0.003	0.005	0.018	0.020	0.018

Notes: This table estimates the effects of alumni networks on fund manager hiring by

$$Hire_{f,m,t} = \alpha_t + \gamma_f + \eta_m + FundAlumni_{f,m} + \epsilon_{f,m,t} \quad (5)$$

fund, manager and year fixed effects are included. The dependent variable in all specifications is an indicator variable denoting that the fund manager m is acquired by mutual fund f in year t . FundAlumni is an indicator variable denoting that the candidate went to the same university as an individual who was a fund manager in the fund at the time of hiring. FamilyAlumni is an indicator variable denoting that the candidate went to the same university as an individual who was a fund manager in the family at the time of hiring. FundAlumni×Bachelor is an indicator variable denoting that the candidate obtained alumni ties through undergraduate programs at the time of hiring. FundAlumni×Master is an indicator variable denoting that the candidate obtained alumni ties through master programs, but excluding MBA programs at the time of hiring. FundAlumni×MBA is an indicator variable denoting that the candidate obtained alumni ties through MBA programs at the time of hiring. FundAlumni×PhD is an indicator variable denoting that the candidate obtained alumni ties through PhD programs at the time of hiring.

Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

Table 13: Propensity Score Matching Estimators

Nearest neighbor is 5				
	No. of obs.	Mean	Diff. (Alumni - NonAlumni)	P-value of diff.
<i>Return</i> _{<i>t</i>+1} (Alumni)	12,142	0.018	-0.015***	0.000
<i>Return</i> _{<i>t</i>+1} (NonAlumni)		0.032	(0.002)	
<i>Alpha</i> _{<i>t</i>+1} (Alumni)	12,142	0.001	0.002***	0.002
<i>Alpha</i> _{<i>t</i>+1} (NonAlumni)		-0.001	(0.001)	
<i>Return</i> _{<i>t</i>+1} + <i>Return</i> _{<i>t</i>+2} (Alumni)	11,651	0.018	-0.023***	0.000
<i>Return</i> _{<i>t</i>+1} + <i>Return</i> _{<i>t</i>+2} (NonAlumni)		0.041	(0.002)	
<i>Alpha</i> _{<i>t</i>+1} + <i>Alpha</i> _{<i>t</i>+2} (Alumni)	11,651	0.002	0.003***	0.000
<i>Alpha</i> _{<i>t</i>+1} + <i>Alpha</i> _{<i>t</i>+2} (NonAlumni)		-0.001	(0.001)	
<i>Tenure</i> (Alumni)	17,784	6.57	-0.870***	0.000
<i>Tenure</i> (NonAlumni)		7.44	(0.095)	
<i>Survive</i> _{<i>t</i>+2} (Alumni)	17,784	0.780	-0.023***	0.005
<i>Survive</i> _{<i>t</i>+2} (NonAlumni)		0.803	(0.008)	
<i>Survive</i> _{<i>t</i>+3} (Alumni)	17,784	0.695	-0.044***	0.000
<i>Survive</i> _{<i>t</i>+3} (NonAlumni)		0.738	(0.009)	
<i>Survive</i> _{<i>t</i>+5} (Alumni)	17,784	0.492	-0.032***	0.001
<i>Survive</i> _{<i>t</i>+5} (NonAlumni)		0.524	(0.010)	
<i>Survive</i> _{<i>t</i>+10} (Alumni)	17,784	0.209	-0.117***	0.000
<i>Survive</i> _{<i>t</i>+10} (NonAlumni)		0.326	(0.008)	

Notes: This table estimates the average treatment effect of treated group by using the propensity score matching algorithm. In this algorithm, the treatment group includes manager acquisition observations of which the acquired manager has alumni ties with the current fund manager. The control group contains manager acquisition observations of which the acquired manager does not have alumni connections with the current fund manager. In the propensity score matching algorithm, we have one-to-five matching of an alumni-acquisition and a non-alumni acquisition. We tested the difference of one-year return (and excess return), two-year return (excess return), tenure, and survival rate after 2, 3, 5, and 10 years of acquisition. Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<i>Tenure</i>	<i>Tenure</i>	<i>Tenure</i>	<i>Survive_{t+2}</i>	<i>Survive_{t+2}</i>	<i>Survive_{t+5}</i>	<i>Survive_{t+10}</i>
<i>FundAlumni</i>	-1.317*** (0.143)	-1.274*** (0.144)	-1.168*** (0.144)	-0.034*** (0.012)	-0.029** (0.014)	-0.057*** (0.016)	-0.117*** (0.014)
<i>Mgr.Ret_{t+1}</i>		2.826*** (0.970)	0.039 (1.067)	0.349*** (0.085)			
<i>Mgr.Flow_{t+1}</i>			1.641*** (0.211)	0.061*** (0.011)			
<i>Mgr.Ret_{t+1} + Mgr.Ret_{t+2}</i>					0.676*** (0.112)	0.224* (0.121)	0.211** (0.099)
<i>Mgr.Flow_{t+1} + Mgr.Flow_{t+2}</i>					-0.000 (0.001)	0.009*** (0.002)	0.010*** (0.002)
Number of obs.	4,203	4,203	4,203	4,203	3,992	3,992	3,992
R-squared	0.020	0.021	0.035	0.013	0.022	0.021	0.044

Note: Significance: * significant at 10%; ** significant at 5%; *** significant at 1%.